

Gopi Maguluri

+1-9255998761 | [Email](#) | [LinkedIn](#) | [Github](#) | [Portfolio](#) | San Francisco (open to relocate)

Data Scientist/ML Engineer with 4+ years of experience developing and deploying end-to-end Machine Learning solutions for business impact.

EDUCATION

University of San Francisco <i>Master of Science in Data Science</i>	San Francisco, California July 2024 - June 2025
BML Munjal University <i>Bachelor of Technology in Electronics and Communication Engineering</i>	New Delhi, India Aug 2017 - June 2021

SKILLS, ACTIVITIES & INTERESTS

Technical Skills: Machine Learning, Deep Learning, Natural Language Processing, Reinforcement Learning, Large Language Models, Prompt Engineering, Generative AI, Statistics, A/B testing, TensorFlow, PyTorch, Shell, Scikit-learn, NumPy, Pandas, SciPy, Python, Linux, SQL, PySpark, API development, FastAPI, Docker, Git, AWS

Certifications & Training: [Deep Generative models](#), Indian Institute of Science, Jan 2023 to May 2023

Activities: President of Performing Arts Club, Data Science Mentor at Sabudh Foundation

EXPERIENCE

ArangoDB <i>Machine Learning Engineer (LLMs, Fine-tuning, Reinforcement Learning, Shell)</i>	San Francisco, California Oct 2024 – Present
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- Developed a **Natural Language to ArangoDB Query Language (AQL) system** by **fine-tuning LLMs** using **PEFT** enabling seamless interaction with graph databases, **improving query efficiency by 78%** for customers.
- Engineered LLM fine-tuning pipelines with **custom reward functions for reinforcement learning**, optimizing GPU utilization and training efficiency, **resulting in a 25% improvement** in AQL generation accuracy.
- Developed an **automated data generation pipeline** for graph databases to fine-tune LLMs, built an **LLM-as-a-Judge based validation system** reducing manual effort by **85%** during my practicum.

Tatras Data <i>Data Scientist (NLP, TensorFlow, PyTorch, Transformers, prompt engineering, FastAPI)</i>	New Delhi, India July 2021 – June 2024
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- Developed an NLP solution to automate matching job categories based on job titles and industries by **training Bi-Encoders, Cross-Encoders and fine-tuning Gemma-2B**, achieving a **75% reduction in matching time**.
- Developed a **production-ready, dockerized API** to interact with the databases in natural language by **automating SQL query generation using LLMs and prompt engineering**, delivering **80% accuracy**.
- Developed and deployed a dockerized end-to-end solution for a text ranking use case using **Word2Vec, FastText, GRUs, BERT, and Sentence Transformers**, **reducing customer's manual effort** in ranking by **90%**.
- Implemented and deployed an **Intelligent Document Processing solution** with document clustering and token classification pipeline by **training LayoutLMv3**, improving customer's document processing **efficiency by 5x**.

Smart Energy Water <i>Data Scientist (Random Forest, XGBoost, Convolutional Neural Networks)</i>	Noida, India Jan 2021 – June 2021
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- Developed a text **classification** solution for customer complaints using **custom-trained FastText embeddings** and an **ensemble of Decision Trees, Random Forest and XGBoost** algorithms, **resulting in 92% accuracy**.
- Accomplished a **research based project** that aimed at **disaggregating household electricity consumption** to the appliance level, involving data analysis, **feature engineering and model training using CNNs**.

PROJECTS

Deep Learning based Chess Bot Github Blog <i>Tech stack: PyTorch, Self-Attention, Squeeze-and-Excitation Networks, Transformers, GPU Cluster</i>	San Francisco, California
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- Secured **1st place** and a **\$10,000 credit** in the **Chess GPU Hackathon** by designing and **training** a high-performance **deep learning based chess model** from scratch on a **48-GPU cluster**.

Fine Tuning Gemma for Law Stack Exchange Github <i>Tech stack: PyTorch, NeMo Curator & Microservices, Data Preprocessing, Large Language Models, Fine-tuning</i>	San Francisco, California
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- Preprocessed 20,000+ legal forum records with **Nvidia NeMo Curator**, and **fine-tuned Gemma 2B model** using NeMo microservices, improving tagging accuracy by 15% during the **ODSC Nvidia Hackathon**.

Raga Identification in Indian Classical Music Github <i>Tech stack: Python, TensorFlow, Data Analysis, Feature Engineering, Convolutional Neural Networks, Flask</i>	New Delhi, India
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- Developed and deployed a **web application** to **predict Ragas from audios** by extracting Mel spectrograms and pitch contours, and **training CNNs**, achieving **82% accuracy** as **Data Science Intern at Sabudh Foundation**.